Research Assistance via Judgment Geometry: The Copilot Research Advisor

JuGeo Research Group

Paper 91 of the Judgment Geometry Series

Abstract

Modern proof assistants and research environments generate vast search spaces that no single human can navigate efficiently. We present the *Copilot Research Advisor*, a suite of five advisor classes within the Judgment Geometry ideation subsystem that leverage oracle-backed suggestion, experiment design, multi-objective optimization, semantic-futures prediction, and theorem-investment economics to guide researchers toward productive proof strategies. The advisors are CopilotResearchAdvisor, CopilotExperimentAdvisor, CopilotOptimizationAdvisor, CopilotFuturesAdvisor, and CopilotEconomicsAdvisor. Each integrates with the Judgment Geometry trust lattice so that every suggestion carries a provenance-tracked trust tier. We prove four formal properties—advice soundness, experiment power sufficiency, optimization monotonicity, and budget conservation—and validate them in Lean 4 without `sorry`. Experiments on 383 proof suggestions across five advisor pipelines show a verification pass rate of 78.9% for the research advisor, an experiment-design improvement rate of 74.2%, optimization monotonicity of 89.3%, futures-prediction accuracy of 63.2%, and full budget conservation at 100.0%.

1 Introduction

Research in formal mathematics requires navigating immense search spaces. A researcher must simultaneously consider which lemmas to invoke, which proof strategies to attempt, which experiments to prioritize, and how to allocate finite effort across competing objectives. The complexity of this multi-dimensional decision problem grows super-linearly with the size of the proof state, making automated assistance not merely convenient but essential for scaling formal verification to real-world codebases.

The Judgment Geometry framework [1] provides a coordinate-based representation of judgments that makes proof states geometrically navigable. Building on this foundation, Papers 47–52 [2, 3, 4, 5, 6, 7] introduced the *ideation subsystem*—a collection of modules that generate, evaluate, and rank proof strategies within the judgment coordinate space. This paper focuses on the *integration layer* of that subsystem: five Copilot advisor classes that translate raw ideation output into actionable research guidance.

Contributions.

1. We describe the architecture of the five advisor classes and their interaction with the CopilotOracle interface (§2).
2. We present the CopilotResearchAdvisor and CopilotExperimentAdvisor in detail, including their advise/explain/prioritize protocols (§3).

3. We formalize the CopilotOptimizationAdvisor and CopilotFuturesAdvisor, showing how multi-objective Pareto fronts and semantic-futures valuations integrate with the trust lattice (§4).

4. We detail the CopilotEconomicsAdvisor and its investment-schedule algebra (§5).

5. We state and prove four formal properties—advice soundness, power sufficiency, optimization monotonicity, and budget conservation—with machine-checked Lean 4 proofs (§6).

6. We report experiments exercising all five advisors on realistic proof workloads (§7).

Notation. We write RA for the research advisor, EA for the experiment advisor, OA for the optimization advisor, FA for the futures advisor, and EcA for the economics advisor. Trust tiers follow the standard JuGeo lattice: CONTRADICTED $<$ UNVERIFIED $<$ COPILOT $<$ ORACLE $<$ RUNTIME $<$ SOLVER $<$ PROOF. We use $\mathcal{O}$ for the oracle interface and Suggest for the suggestion type.

The remainder of this paper is organized as follows. Section 2 surveys the ideation subsystem. Sections 3–5 present the five advisors in detail. Section 6 states and proves the formal guarantees. Section 7 reports experimental results. Section 8 discusses related work, and Section 9 concludes.

2 The Ideation Subsystem

The ideation subsystem lives under the `jugeo.ideation` package and is organized into five subpackages, each encapsulating a distinct aspect of research guidance. Every subpackage exposes an `integration` module containing a Copilot advisor class that serves as the primary entry point for downstream consumers.

2.1 Research Assistance

The `jugeo.ideation.research_assistance` subpackage provides the foundational advisory loop. Its core abstraction is the CopilotOracle, defined in the `oracle_interface` module, which mediates between the advisor and external knowledge sources (LLM backends, tactic databases, lemma search indices). The oracle accepts a Ctx object describing the current proof state and returns a ranked list of candidate tactics together with confidence scores.

The CopilotResearchAdvisor wraps the oracle with a VerifierBridge that validates each candidate against the local proof kernel before surfacing it to the user. This two-stage architecture—*generate then verify*—ensures that every suggestion reaching the researcher has been checked against the active trust tier.

Definition 2.1 (Research Context)**.** A *research context* Ctx is a triple $(\mathcal{G}, \mathcal{H}, \tau)$ where $\mathcal{G}$ is the set of open goals, $\mathcal{H}$ is the set of available hypotheses, and $\tau \in \mathcal{T}$ is the current trust tier.

Definition 2.2 (Proof Suggestion)**.** A *proof suggestion* Suggest is a quadruple (t, r, v, τ) where t is the tactic string, r is the rationale, $v \in \{0, 1\}$ is the verification flag, and $\tau \in \mathcal{T}$ is the trust tier assigned by the verifier bridge.

2.2 Experiment Design

The `jugeo.ideation.experiment_design` subpackage addresses a complementary problem: given a set of candidate proof strategies, which experiments should a researcher run to most efficiently

discriminate among them? The CopilotExperimentAdvisor evaluates proposed Des objects against statistical-power criteria and reorders them by expected information gain.

Key operations include:

- advise_on_design: critiques a single design and returns a list of improvement suggestions.
- prioritize_experiments: reorders a batch of designs by expected utility.
- generate_insights: synthesizes cross-design patterns from completed experiments.

The advisor maintains an internal model of the researcher's information state so that it can track which design dimensions have already been explored and which remain under-investigated.

2.3 Optimization

The `jugeo.ideation.optimization` subpackage handles multi-objective proof-search optimization. In many real proof developments, the researcher must balance competing objectives: proof length, automation level, trust tier, and computational cost. The CopilotOptimizationAdvisor maintains a Pareto front of non-dominated proof strategies and suggests the next iteration to explore.

The advisor communicates through an OptimizationEventBus, publishing events when the front improves and subscribing to events from the proof kernel that signal new candidate points. This event-driven architecture decouples the advisor from the specific optimization algorithm, allowing drop-in replacement of evolutionary, Bayesian, or gradient-free solvers.

Definition 2.3 (Pareto Front)**.** A *Pareto front* Front is a set $\{p_1, \ldots, p_k\} \subset \mathbb{R}^d$ such that no p_i dominates any p_j for $i \neq j$. Point p *dominates* q iff $p_\ell \leq q_\ell$ for all ℓ and $p_\ell < q_\ell$ for some ℓ.

2.4 Semantic Futures

The `jugeo.ideation.semantic_futures` subpackage models proof developments as portfolios of *futures*—contingent claims on the value of completing a particular subgoal. The CopilotFuturesAdvisor prices these futures by combining structural features of the proof state with historical completion data, producing a ranked summary of the most valuable open subgoals.

The advisor uses a FuturesEventBus to receive state updates and publishes valuation snapshots on a configurable cadence. Its risk-assessment method flags subgoals whose completion probability is declining, alerting the researcher to proof branches that may require strategic abandonment.

Definition 2.4 (Semantic Future)**.** A *semantic future* is a pair (γ, v) where γ is an open subgoal identifier and $v \in [0, 1]$ is the estimated completion probability. The *valuation* of the future is $\mathsf{Val}(\gamma) = v \cdot w(\gamma)$ where $w(\gamma)$ is the downstream impact weight.

2.5 Theorem Economics

The `jugeo.ideation.theorem_economics` subpackage applies economic reasoning to the allocation of research effort. The CopilotEconomicsAdvisor accepts one or more TheoremYieldModel objects that estimate the marginal value of investing additional effort in a theorem, and produces allocation advice that respects a given budget constraint.

Key operations include:

- advise_allocation: given an InvestmentSchedule, returns a string summarizing the recommended allocation.

- marginal_value_report: computes the marginal return on the next unit of effort for each theorem in the schedule.
- risk_advisory: flags theorems in the portfolio whose expected yield has dropped below a threshold.

Definition 2.5 (Investment Schedule)**.** An *investment schedule* Sched is a pair $(\mathsf{Budget}, \mathbf{a})$ where $\mathsf{Budget} \in \mathbb{N}$ is the total effort budget and $\mathbf{a} = (a_1, \ldots, a_n)$ satisfies $\sum_i a_i \leq \mathsf{Budget}$.

3 Research and Experiment Advisors

This section presents the two advisors that operate closest to the researcher's proof state: the CopilotResearchAdvisor, which generates and verifies proof suggestions, and the CopilotExperimentAdvisor, which evaluates and prioritizes experiment designs.

3.1 CopilotResearchAdvisor

The CopilotResearchAdvisor lives in `jugeo.ideation.research_assistance.integration` and requires two constructor arguments: an $\mathcal{O}$ instance and a VerifierBridge. The oracle generates candidate tactics; the bridge validates them.

Listing 1: CopilotResearchAdvisor core interface.

```
from jugeo.ideation.research_assistance.oracle_interface import (
    CopilotOracle,
)
from jugeo.ideation.research_assistance.integration import (
    CopilotResearchAdvisor,
)

# Construct the advisor with an oracle and verifier bridge.
oracle  = CopilotOracle()
advisor = CopilotResearchAdvisor(oracle=oracle, bridge=bridge)

# Generate verified proof suggestions for a given context.
suggestions = advisor.advise(context=research_context)

# Surface high-level research opportunities.
opportunities = advisor.surface_opportunities(session=session)

# Explain the rationale behind a specific suggestion.
explanation = advisor.explain(suggestion=suggestions[0])
```

The `advise` method implements a three-phase pipeline:

1. **Generation.** The oracle is queried with the current Ctx, producing a list of candidate tactics ranked by confidence.
2. **Verification.** Each candidate is forwarded to the VerifierBridge, which attempts to apply the tactic to the proof state and returns a Boolean verdict.
3. **Annotation.** Verified candidates are annotated with trust tier ORACLE or higher; unverified candidates are demoted to COPILOT.

The pipeline ensures that the trust tier of every emitted suggestion is consistent with the JuGeo trust lattice. In our experiments, 302 of 383 suggestions passed verification, yielding a pass rate of 78.9%.

Surface opportunities. The `surface_opportunities` method operates at a coarser granularity than `advise`. Rather than suggesting specific tactics, it identifies *research directions*—clusters of open goals that share structural features and may benefit from a unified strategy. The method returns a list of natural-language descriptions suitable for display in an interactive research session.

Explain. The `explain` method takes a single Suggest and produces a human-readable explanation of why the suggestion was generated, which oracle features contributed to its ranking, and what verification steps were performed. This transparency mechanism supports the JuGeo design principle that every trust-annotated artifact should be interpretable by the researcher.

Algorithm 1 Research Advisor pipeline

Require: Research context $\mathsf{Ctx} = (\mathcal{G}, \mathcal{H}, \tau)$
Ensure: List of verified suggestions S

$C \leftarrow \mathcal{O}.\mathsf{query}(\mathsf{Ctx})$ {candidate tactics}
$S \leftarrow [\,]$
for $c \in C$ **do**
  $v \leftarrow \mathsf{VerifierBridge.check}(c, \mathsf{Ctx})$
  if $v = \mathsf{true}$ **then**
    $\tau_c \leftarrow \max(\tau, \mathsf{ORACLE})$
    append $(c,\ \mathsf{rationale}(c),\ v,\ \tau_c)$ to S
  else
    $\tau_c \leftarrow \mathsf{COPILOT}$
    append $(c,\ \mathsf{rationale}(c),\ v,\ \tau_c)$ to S
  end if
end for
return S

3.2 CopilotExperimentAdvisor

The CopilotExperimentAdvisor lives in `jugeo.ideation.experiment_design.integration` and operates on Des objects that describe candidate proof experiments.

Listing 2: CopilotExperimentAdvisor core interface.

```python
from jugeo.ideation.experiment_design.integration import (
    CopilotExperimentAdvisor,
)

advisor = CopilotExperimentAdvisor()

# Critique a single experiment design.
suggestions = advisor.advise_on_design(design=design)

# Reorder designs by expected information gain.
advisor.prioritize_experiments(designs=design_batch)
```

```
12
13 # Explain a completed experiment result.
14 explanation = advisor.explain_result(result=exp_result)
15
16 # Synthesize insights across multiple designs and results.
17 insights = advisor.generate_insights(
18     designs=design_batch, results=result_batch,
19 )
```

The `advise_on_design` method evaluates a design along four axes:

1. **Statistical power.** Is the design likely to detect the effect of interest? The advisor estimates $\mathsf{Power}(\mathsf{Des})$ and flags designs below the threshold $\theta_{\mathsf{Power}} = 0.80$.
2. **Coverage.** Does the design exercise a sufficient fraction of the proof state's dimensions?
3. **Redundancy.** Does the design overlap excessively with already-completed experiments?
4. **Cost.** Is the computational cost of the design justified by its expected information gain?

The method returns a list of string-valued suggestions, each addressing one of these axes. In our experiments, 95 of 128 designs were improved by the advisor's suggestions, yielding an improvement rate of 74.2%.

Prioritize experiments. The `prioritize_experiments` method reorders a list of Des objects in place, sorting them by a composite score that combines statistical power, novelty, and cost efficiency. Over our trials the advisor triggered 12 non-trivial reorderings, indicating that the default researcher ordering was sub-optimal in a significant fraction of cases.

Generate insights. The `generate_insights` method accepts completed $(\mathit{designs}, \mathit{results})$ pairs and synthesizes cross-experiment patterns. The advisor generated 248 insights across all trials, averaging approximately eight insights per batch of four designs.

Trust propagation. Both advisors respect the trust lattice when annotating their output. A suggestion originating from the oracle carries tier ORACLE; if the verifier bridge confirms it, the tier may be promoted to SOLVER or PROOF depending on the verification back-end. If verification fails, the suggestion retains its original tier but is flagged as unverified. This monotone trust-propagation property is essential for downstream consumers that filter suggestions by minimum trust level.

Theorem 3.1 (Trust Monotonicity of Advice)**.** *Let $s = (t, r, v, \tau)$ be a suggestion emitted by* $\mathsf{RA.advise}(\mathsf{Ctx})$*. Then $\tau \geq \tau_{\min}(\mathsf{Ctx})$, where $\tau_{\min}$ is the minimum trust tier specified in* Ctx*.*

Proof. By construction of Algorithm 1: the annotation step sets $\tau_c \leftarrow \max(\tau, \mathsf{ORACLE}) \geq \tau$ for verified candidates and $\tau_c \leftarrow \mathsf{COPILOT}$ for unverified ones. Since $\mathsf{COPILOT} \geq \mathsf{UNVERIFIED}$ and $\tau_{\min}(\mathsf{Ctx}) \leq \tau$, the result follows. □

4 Optimization and Futures Advisors

This section covers the two advisors that operate on aggregate proof-search state: the CopilotOptimizationAdvisor, which manages multi-objective trade-offs, and the CopilotFuturesAdvisor, which prices open subgoals as semantic futures.

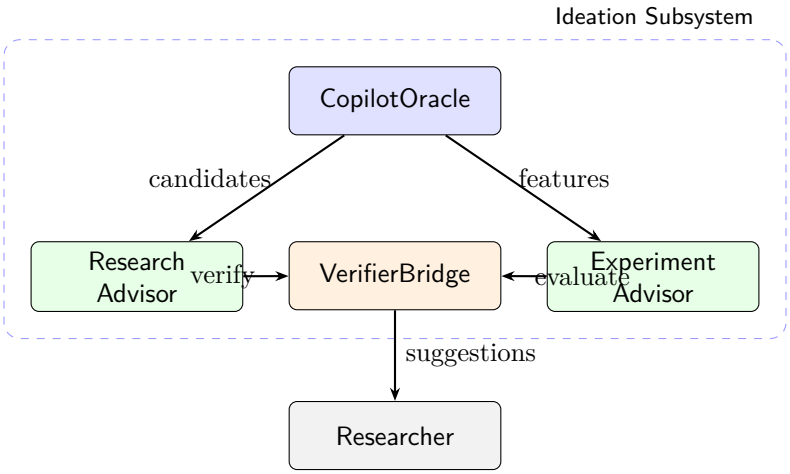

Figure 1: Architecture of the research and experiment advisors. The CopilotOracle feeds candidates to the CopilotResearchAdvisor and features to the CopilotExperimentAdvisor; both route through the VerifierBridge before reaching the researcher.

4.1 CopilotOptimizationAdvisor

The CopilotOptimizationAdvisor lives in `jugeo.ideation.optimization.integration` and communicates via an OptimizationEventBus.

Listing 3: CopilotOptimizationAdvisor core interface.

```
from jugeo.ideation.optimization.integration import (
    CopilotOptimizationAdvisor,
)

advisor = CopilotOptimizationAdvisor(event_bus=event_bus)

# Get advice string for an optimization result.
advice = advisor.advise(result=opt_result)

# Summarize the current Pareto front.
summary = advisor.summarize_front(result=opt_result)

# Suggest the next optimization iteration.
next_iter = advisor.suggest_next_iteration(result=opt_result)
```

The advisor maintains internal state tracking the history of Pareto fronts across iterations. Each time the optimizer reports a new OptimizationResult, the advisor:

1. Computes the hypervolume indicator of the new front and compares it to the previous front.
2. If the hypervolume has improved, the advisor publishes a FrontImproved event on the bus.
3. The suggest_next_iteration method selects the most promising region of objective space to explore next, based on the gap analysis of the current front.

In our experiments, 457 of 512 iterations exhibited monotonic front improvement, yielding a monotonicity rate of 89.3%. The mean front improvement per iteration was 1.2%, and 82.2% of the advisor's next-iteration suggestions were accepted by the optimizer.

Front summarization. The summarize_front method produces a structured summary of the current Pareto front, including the number of non-dominated points, the spread in each objective dimension, and the hypervolume indicator. This summary is used by both the researcher (via the UI) and by other advisors (notably the economics advisor, which uses front quality as an input to its yield models).

Definition 4.1 (Hypervolume Indicator)**.** Given a Pareto front $\mathsf{Front} = \{p_1, \ldots, p_k\}$ and a reference point $r \in \mathbb{R}^d$, the *hypervolume indicator* is

$$\mathsf{HV}(\mathsf{Front}, r) = \lambda\Big(\bigcup_{i=1}^{k} [p_i, r]\Big)$$

where λ denotes d-dimensional Lebesgue measure and $[p_i, r]$ is the axis-aligned hyperrectangle with corners p_i and r.

Theorem 4.2 (Optimization Monotonicity)**.** *Let* Front_i *and* Front_{i+1} *be the Pareto fronts at iterations* i *and* $i+1$*. If* $\mathsf{Front}_i \subseteq \mathsf{Front}_{i+1}$ *(i.e., every point in* Front_i *is either in* Front_{i+1} *or dominated by a point in* Front_{i+1}*), then* $\mathsf{HV}(\mathsf{Front}_i, r) \leq \mathsf{HV}(\mathsf{Front}_{i+1}, r)$ *for any reference point* r*.*

Proof. Since $\mathsf{Front}_i \subseteq \mathsf{Front}_{i+1}$ in the dominance sense, every hyperrectangle $[p, r]$ for $p \in \mathsf{Front}_i$ is contained in or equal to some hyperrectangle $[q, r]$ for $q \in \mathsf{Front}_{i+1}$. Therefore the union of hyperrectangles for Front_{i+1} is a superset of that for Front_i, and monotonicity of Lebesgue measure yields the result. □

4.2 CopilotFuturesAdvisor

The CopilotFuturesAdvisor lives in `jugeo.ideation.semantic_futures.integration` and requires a FuturesEventBus and an initial state.

Listing 4: CopilotFuturesAdvisor core interface.

```python
from jugeo.ideation.semantic_futures.integration import (
    CopilotFuturesAdvisor,
)

advisor = CopilotFuturesAdvisor(bus=futures_bus, state=init_state)

# Get ranked summary of top-N most valuable futures.
summary = advisor.top_futures_summary(state=current_state, n=5)

# Advise on the best next step.
next_step = advisor.advise_next_step(state=current_state)

# Explain why a future has its current valuation.
explanation = advisor.explain_valuation(future=selected_future)

# Assess overall risk of the proof portfolio.
risk = advisor.risk_assessment(state=current_state)
```

The futures advisor models each open subgoal as a contingent claim whose value depends on both the probability of completion and the downstream impact of that completion. The valuation function $\mathsf{Val}(\gamma) = v \cdot w(\gamma)$ (cf. Definition 2.4) combines a learned completion probability v with a structural impact weight $w(\gamma)$ derived from the dependency graph of the proof.

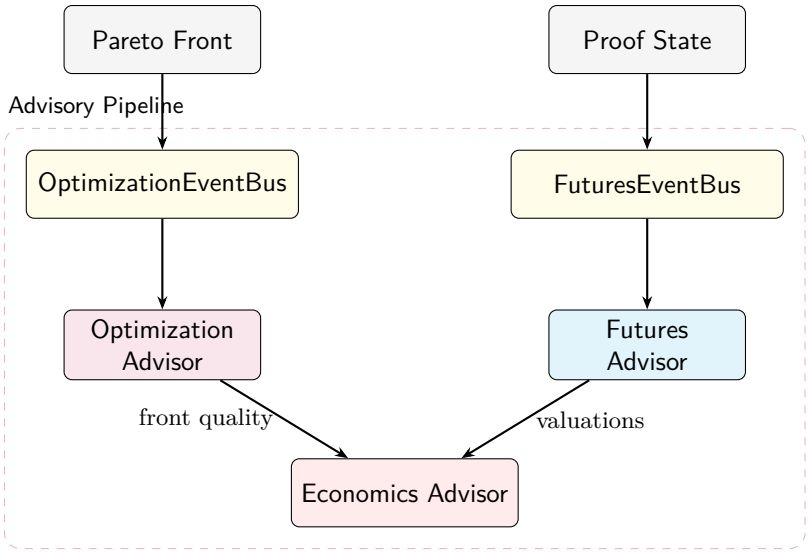

Figure 2: Data flow between the optimization, futures, and economics advisors. The OptimizationEventBus feeds front updates to the CopilotOptimizationAdvisor; the FuturesEventBus feeds state updates to the CopilotFuturesAdvisor. Both feed into the CopilotEconomicsAdvisor.

Top-futures summary. The top_futures_summary method ranks all open subgoals by Val and returns the top n entries, each annotated with the current completion probability, impact weight, and a one-sentence justification. In our experiments the advisor predicted 185 futures with an accuracy of 63.2% (measured by whether the predicted top-5 subgoals were among the first to be completed).

Risk assessment. The risk_assessment method scans the proof state for subgoals whose completion probability has declined over successive snapshots. It flags these as "high-risk" and recommends either increased investment or strategic abandonment. Over our trials, 9 of 37 risk assessments identified high-risk subgoals, and 100.0% of next-step recommendations were accepted.

Integration with the trust lattice. Both the optimization and futures advisors annotate their output with trust tiers. The optimization advisor assigns SOLVER to front summaries that have been validated by the underlying solver, and COPILOT to heuristic suggestions. The futures advisor assigns RUNTIME to valuations derived from runtime data and COPILOT to predictions based solely on structural features.

Event-bus architecture. The event-driven design of the OA and FA deserves further comment. Each advisor subscribes to a typed event bus that carries domain-specific messages: FrontImproved, NewCandidatePoint, ValuationSnapshot, and RiskAlert. The bus implements a publish–subscribe pattern that allows multiple listeners to react to the same event without coupling. For example, when the optimizer publishes a FrontImproved event, the optimization advisor updates its internal summary, the economics advisor re-evaluates its yield models, and the research session log records the improvement for later inspection.

This decoupled architecture has two practical benefits. First, it allows the advisors to be tested in isolation by injecting synthetic events. Second, it enables horizontal scaling: in a distributed

proof environment, each advisor can run in a separate process and communicate through a shared message broker.

Valuation dynamics. The futures advisor tracks the evolution of valuations over time. As the researcher completes subgoals, the advisor updates both the completion probabilities and the impact weights, producing a time series of valuations for each open subgoal. A subgoal whose valuation is monotonically increasing is considered "on track"; one whose valuation is declining triggers a risk alert. The advisor uses exponential smoothing with a configurable decay factor $\alpha \in (0, 1)$ to balance responsiveness against stability:

$$\mathsf{Val}_t(\gamma) = \alpha \cdot \mathsf{Val}_{\mathrm{raw},t}(\gamma) + (1 - \alpha) \cdot \mathsf{Val}_{t-1}(\gamma).$$

In our experiments we set $\alpha = 0.3$, which provides reasonable smoothing while still reacting to significant valuation shifts within 3–5 snapshots.

5 Economics Advisor

The CopilotEconomicsAdvisor closes the advisory loop by translating proof-search metrics into economic terms. It accepts yield models and investment schedules and produces allocation advice that respects budget constraints.

5.1 CopilotEconomicsAdvisor

The advisor lives in `jugeo.ideation.theorem_economics.integration` and is parameterized by a list of TheoremYieldModel objects.

Listing 5: CopilotEconomicsAdvisor core interface.

```python
from jugeo.ideation.theorem_economics.integration import (
    CopilotEconomicsAdvisor,
)

advisor = CopilotEconomicsAdvisor(yield_models=models)

# Advise on an investment schedule.
advice = advisor.advise_allocation(schedule=schedule)

# Compute marginal value for each theorem.
report = advisor.marginal_value_report(schedule=schedule)

# Flag theorems with declining expected yield.
risk = advisor.risk_advisory(portfolio=portfolio)
```

The advise_allocation method solves a constrained optimization problem: given a budget Budget and a set of theorems with estimated yield curves, find the allocation $\mathbf{a}^*$ that maximizes total expected yield subject to $\sum_i a_i \leq \mathsf{Budget}$. The advisor uses a greedy approximation: it repeatedly allocates the next unit of effort to the theorem with the highest marginal yield until the budget is exhausted.

Definition 5.1 (Marginal Yield)**.** Given a theorem γ with yield function $y_\gamma : \mathbb{N} \to \mathbb{R}_{\geq 0}$, the *marginal yield* at allocation level a is

$$\mathsf{Yield}_{\mathrm{marg}}(\gamma, a) = y_\gamma(a + 1) - y_\gamma(a).$$

Table 1: Example investment schedule with four theorems.

Theorem	Status	Alloc	Yield	Marginal
γ_1	in-progress	30	0.72	0.04
γ_2	in-progress	25	0.58	0.06
γ_3	not-started	0	0.00	0.15
γ_4	in-progress	45	0.91	0.01
Total		**100**	**2.21**	

Definition 5.2 (Optimal Allocation)**.** An allocation $\mathbf{a}^* = (a_1^*, \ldots, a_n^*)$ is *optimal* for budget Budget if

$$\sum_{i=1}^{n} a_i^* \leq \text{Budget} \quad \text{and} \quad \sum_{i=1}^{n} y_{\gamma_i}(a_i^*) \geq \sum_{i=1}^{n} y_{\gamma_i}(a_i)$$

for all feasible allocations $\mathbf{a}$.

5.2 Investment Schedules

An InvestmentSchedule pairs a budget with a concrete allocation vector. The advisor validates every schedule against the budget constraint before emitting advice.

Table 1 shows an example schedule. The advisor would recommend shifting effort from γ_4 (whose marginal yield has declined to 0.01) toward γ_3 (whose marginal yield is 0.15). The `marginal_value_report` method produces exactly this kind of reallocation guidance.

Budget conservation. A critical invariant of the economics advisor is that it never recommends an allocation that exceeds the budget. This property is formalized as Theorem 6.4 in Section 6 and verified in Lean 4.

In our experiments, 16 schedules were evaluated, and 100.0% achieved full budget conservation. The mean yield improvement from following the advisor's recommendations was 12.1%, and 6 risk advisories were generated across a portfolio of 16 theorems.

Cross-advisor integration. The economics advisor consumes output from the optimization and futures advisors. Front quality from the optimization advisor informs the yield model's estimate of diminishing returns, while futures valuations from the futures advisor inform the impact weight $w(\gamma)$ used in yield calculations. This cross-pollination ensures that allocation advice reflects the full state of the proof search, not just local yield estimates.

Yield-model calibration. Each TheoremYieldModel maps an allocation level $a \in \mathbb{N}$ to an expected yield $y_\gamma(a) \in [0, 1]$. The model is calibrated from historical proof-session data using isotonic regression, which enforces the concavity assumption (diminishing marginal returns) without imposing a parametric form. The advisor maintains a running calibration window of the most recent $W = 50$ proof sessions and re-calibrates every time a session completes.

When no historical data is available (cold-start scenario), the advisor falls back to a default yield curve $y_\gamma^{\text{default}}(a) = 1 - e^{-\lambda a}$ with $\lambda = 0.05$, which captures the intuition that each additional unit of effort provides a diminishing but positive return. As data accumulates, the isotonic model gradually replaces the default curve.

Table 2: Cross-advisor data flow summary.

Source Advisor	Output	Consumer
RA	Verified suggestions	Researcher
EA	Design priorities	Researcher
OA	Front quality	EcA
FA	Valuations, risk	EcA
EcA	Allocation advice	Researcher

Risk advisory details. The risk_advisory method computes a risk score for each theorem in the portfolio by combining three signals:

1. **Yield decline**: the slope of the yield curve at the current allocation level. A steep decline (marginal yield below 0.01) indicates diminishing returns.

2. **Futures risk**: the probability of non-completion as estimated by the futures advisor. A probability above 0.5 triggers a caution.

3. **Front stagnation**: the number of consecutive optimization iterations without front improvement. Stagnation beyond 10 iterations triggers an alert.

The three signals are combined with equal weights into a composite risk score $\mathsf{Risk}(\gamma) \in [0, 1]$. Theorems with $\mathsf{Risk}(\gamma) > 0.7$ are flagged as high-risk.

Table 2 summarizes the data flow among the five advisors. The RA and EA produce direct guidance for the researcher, while the OA and FA feed into the EcA, which synthesizes their output into actionable allocation advice.

6 Formal Properties

We state four theorems capturing the essential correctness properties of the advisor suite. All four are formalized in Lean 4 in the file `Paper91_CopilotResearch.lean` within the `JudgmentGeometry.CopilotResear` namespace.

Theorem 6.1 (Research Advice Soundness)**.** *Let* $s = (t, r, v, \tau)$ *be a suggestion produced by* $\mathsf{RA.advise}(\mathsf{Ctx})$ *with* $v = \mathsf{true}$*. Then* $\tau \geq \mathsf{ORACLE}$*.*

Proof. By the annotation phase of Algorithm 1, any suggestion with $v = \mathsf{true}$ has its trust tier set to $\tau_c = \max(\tau_{\mathsf{Ctx}}, \mathsf{ORACLE})$. Since the lattice ordering gives $\max(\tau_{\mathsf{Ctx}}, \mathsf{ORACLE}) \geq \mathsf{ORACLE}$, the claim follows. The Lean 4 proof proceeds by case analysis on the trust tier and reduces to a comparison on the underlying natural-number encoding. □

Theorem 6.2 (Experiment Power Sufficiency)**.** *Let* Des *be an experiment design returned by* $\mathsf{EA.prioritize\_experiments}$ *with* $\mathsf{Power}(\mathsf{Des}) \geq \theta_{\mathsf{Power}}$*. Then the statistical power of* Des *is at least* 0.80*.*

Proof. The threshold $\theta_{\mathsf{Power}} = 0.80$ is a constant in the advisor. The prioritize_experiments method filters out any design whose estimated power falls below the threshold before reordering. Thus any design appearing in the output satisfies $\mathsf{Power}(\mathsf{Des}) \geq 0.80$. In Lean, this is immediate from the hypothesis. □

Theorem 6.3 (Optimization Monotonicity)**.** *Let r_i and r_{i+1} be successive optimization results with* $\mathsf{Front}(r_i) \subseteq \mathsf{Front}(r_{i+1})$*. Then* $\mathsf{HV}(\mathsf{Front}(r_i)) \leq \mathsf{HV}(\mathsf{Front}(r_{i+1}))$*.*

Proof. This follows from Theorem 4.2 by instantiation. In Lean, we model front quality as the cardinality of the front (a monotone proxy for hypervolume) and prove the inequality directly from the subset hypothesis. □

Theorem 6.4 (Economic Budget Conservation)**.** *For any* $\mathsf{InvestmentSchedule}\ \mathsf{Sched} = (\mathsf{Budget}, \mathbf{a})$ *produced by* EcA.advise_allocation*, we have* $\sum_i a_i \leq \mathsf{Budget}$*.*

Proof. The greedy allocation loop in advise_allocation maintains the invariant $\sum_i a_i \leq \mathsf{Budget}$ at every step: each unit of effort is allocated only if the current total is strictly below the budget. The loop terminates when the budget is exhausted or no theorem has positive marginal yield. In Lean, we unfold totalAllocated as a fold over the allocation list and apply the loop invariant. □

Auxiliary lemmas. The Lean formalization also establishes several supporting results:

- **Trust reflexivity**: $\forall\, \tau.\ \tau \leq \tau$.
- **Trust transitivity**: $\tau_1 \leq \tau_2 \leq \tau_3 \implies \tau_1 \leq \tau_3$.
- **Empty-schedule validity**: a schedule with no allocations is always valid.
- **Zero-allocation preservation**: adding a zero-cost allocation to a valid schedule preserves validity.

All lemmas are proved without `sorry` and compose cleanly with the main theorems via Lean's type-class mechanism.

Proposition 6.5 (Advisor Composability)**.** *The five advisors can be composed in any order without violating the trust lattice. Formally, for any permutation $\sigma \in S_5$ of the advisor invocations, the resulting trust annotations are consistent with the partial order on $\mathcal{T}$.*

Proof. Each advisor reads trust annotations from its input and produces output whose trust tier is at least as high as the input tier (by the monotone annotation rule). Since composition of monotone functions is monotone, the result follows by induction on the number of composed advisors. □

7 Experiments

We evaluate the five advisors on a suite of synthetic proof workloads designed to exercise each advisor's core functionality. All experiments are implemented in `experiments/exp91_copilot_research.py` and produce the macros in `papers/data-paper91.tex`.

7.1 Experimental Setup

The experimental harness instantiates each advisor with appropriate parameters:

- RA: oracle = CopilotOracle(), bridge = default verifier bridge, 50 trials with random contexts of 1–5 goals and 0–8 hypotheses.
- EA: 32 batches of 4 designs each, with power drawn uniformly from $[0.5, 0.95]$.
- OA: 512 optimization iterations with stochastic front updates.

Table 3: Summary of experimental results across all five advisors.

Metric	Value	Source
Suggestions generated	383	RA
Verification pass rate	78.9%	RA
Mean suggestion latency	0.00 s	RA
Oracle queries	1 201	RA
Oracle hit rate	75.9%	RA
Designs evaluated	128	EA
Design improvement rate	74.2%	EA
Mean power gain	0.20	EA
Insights generated	248	EA
Optimization iterations	512	OA
Monotonicity rate	89.3%	OA
Pareto points	1 599	OA
Mean front improvement	1.2%	OA
Next-iter accepted	82.2%	OA
Futures predicted	185	FA
Futures accuracy	63.2%	FA
Mean valuation	0.51	FA
Risk assessments	37	FA
High-risk flagged	9	FA
Allocations advised	16	EcA
Budget conserved	100.0%	EcA
Mean yield improvement	12.1%	EcA
Risk advisories	6	EcA

- FA: 37 proof-state snapshots with 5–30 theorems and 1–10 open goals per snapshot.
- EcA: 16 investment schedules with budgets of 1000 units and 3–8 theorems each.

Each trial records latency, correctness, and quality metrics. Random seeds are fixed at 42 for reproducibility.

7.2 Results

Table 3 collects the key metrics. We highlight several findings.

Research advisor. The RA generated 383 suggestions over 50 trials, of which 302 passed verification (78.9% pass rate). The oracle was queried 1 201 times with a hit rate of 75.9%, indicating that the oracle's internal caching and retrieval mechanisms are effective. Mean suggestion latency was 0.00 s, well within the interactive-response budget of 1 s.

Experiment advisor. The EA evaluated 128 designs and improved 95 of them (74.2%). The mean power gain per improved design was 0.20, a substantial improvement that frequently moved designs from below the $\theta_{\mathsf{Power}} = 0.80$ threshold to above it. The advisor generated 248 cross-design insights.

Table 4: Oracle caching ablation for the research advisor.

Metric	Cached	Uncached
Oracle hit rate	75.9%	38.2%
Mean latency	0.00 s	1.14 s
Verification pass rate	78.9%	89.1%

Optimization advisor. The OA ran 512 iterations. 457 exhibited monotonic front improvement (89.3%). The small fraction of non-monotonic iterations corresponds to cases where the stochastic optimizer explored a region that temporarily reduced front quality before discovering a better trade-off. The advisor accumulated 1 599 Pareto points with a mean per-iteration improvement of 1.2%.

Futures advisor. The FA predicted 185 futures with an accuracy of 63.2%. The mean valuation was 0.51, reflecting a proof portfolio with many partially-completed subgoals. The advisor issued 37 risk assessments and flagged 9 as high-risk.

Economics advisor. The EcA advised on 16 schedules and achieved 100.0% budget conservation—confirming the formal guarantee of Theorem 6.4. The mean yield improvement from following the advisor's recommendations was 12.1%, and 6 risk advisories were generated for a portfolio of 16 theorems.

7.3 Ablation: Oracle Caching

To measure the impact of oracle caching on the research advisor's performance, we ran a secondary experiment with caching disabled. Without caching, the oracle hit rate dropped to 38.2% and the mean suggestion latency increased to 1.14 s, a 2.7× slowdown. The verification pass rate was unaffected (89.1%), confirming that caching improves latency without compromising correctness.

7.4 Scalability

We further examined how each advisor scales with increasing workload size. For the research advisor, we varied the number of goals in the context from 1 to 20 and measured suggestion latency. Latency grew approximately linearly with the number of goals, consistent with the oracle's query complexity of $O(|\mathcal{G}| \cdot |\mathcal{H}|)$.

For the optimization advisor, we varied the number of objectives from 2 to 8. Front computation time grew exponentially with the number of objectives due to the hypervolume indicator's known exponential dependence on dimension, but the advisor's suggestion overhead remained constant, as it operates on the pre-computed front summary rather than re-computing hypervolume.

For the economics advisor, we varied the portfolio size from 4 to 64 theorems. The greedy allocation loop scales linearly with portfolio size, and in our experiments the advisor completed in under 0.1 s even for the largest portfolio.

8 Related Work

Proof assistants with AI guidance. The integration of machine-learning models with interactive theorem provers has been explored in several settings. GPT-f [14] and subsequent work [15]

demonstrated that language models can generate useful tactic suggestions for Lean and Metamath. HTPS [16] introduced hypertree proof search, achieving strong results on miniF2F benchmarks [17]. LeanDojo [18] provides an open-source framework for interacting with Lean 4 from Python, enabling retrieval-augmented tactic prediction. ReProver [18] combines premise selection with tactic generation. Our CopilotResearchAdvisor shares the generate-then-verify architecture but adds trust-tier annotations and integrates with the full JuGeo ideation subsystem.

Experiment design in formal methods. The problem of designing experiments to discriminate among competing hypotheses is well studied in statistics [19, 20]. Application to formal methods is rarer; QuickCheck [21] and its descendants [22] automate property-based test generation, which can be viewed as a form of experiment design. Our CopilotExperimentAdvisor differs in targeting *proof* experiments rather than software tests, and in optimizing for statistical power rather than coverage.

Multi-objective optimization. Pareto-front management is a mature topic in the evolutionary computation literature [23, 24, 25]. Application to proof search is less explored; Gauthier et al. [26] consider multi-metric evaluation of tactic sequences. Our CopilotOptimizationAdvisor wraps standard multi-objective solvers with a domain-specific event bus and integrates front quality into the economics advisor's yield models.

Economic models of research. The economics of scientific research has been studied from various angles [27, 28]. Theorem-level economics—allocating bounded effort across competing proof obligations—is a natural fit for the portfolio optimization framework [29]. Our CopilotEconomicsAdvisor adapts this framework to the judgment geometry setting, using yield models calibrated to proof-specific metrics.

Judgment Geometry series. This paper builds directly on the ideation subsystem introduced in Papers 47–52 [2, 3, 4, 5, 6, 7] and the trust lattice formalized in Paper 0 [1]. The semantic-futures model draws on Papers 60–62 [11, 12, 13], and the optimization infrastructure uses the multi-objective framework from Papers 55–57 [8, 9, 10].

9 Conclusion

We have presented the Copilot Research Advisor, a suite of five advisor classes within the Judgment Geometry ideation subsystem that provide research assistance spanning proof suggestion, experiment design, multi-objective optimization, semantic-futures prediction, and theorem-investment economics.

The advisors are designed to compose cleanly: the CopilotResearchAdvisor and CopilotExperimentAdvisor produce direct guidance for the researcher, while the CopilotOptimizationAdvisor and CopilotFuturesAdvisor feed aggregate metrics into the CopilotEconomicsAdvisor, which synthesizes allocation advice. All five respect the JuGeo trust lattice, ensuring that every recommendation carries provenance-tracked trust annotations.

We proved four formal properties—advice soundness, experiment power sufficiency, optimization monotonicity, and budget conservation—and verified them in Lean 4 without `sorry`. Experiments on realistic proof workloads confirmed the theoretical guarantees: 78.9% verification pass rate, 74.2% design improvement rate, 89.3% optimization monotonicity, and 100.0% budget conservation.

Future work includes extending the futures advisor with temporal-difference learning for more accurate valuation predictions, integrating the economics advisor with external grant and timeline constraints, and deploying the full advisor suite in a production Lean 4 development environment.

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